

ISLAMIC UNIVERSITY OF TECHNOLOGY



MACHINE LEARNING LAB

CSE 4622

Lab 1: Introduction to Python, NumPy, and Plotting

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1 Introduction to Google Colab

Google Colab is a cloud-based Python environment. It allows you to write and execute Python code in your browser.

Steps to Get Started

1. Go to <https://colab.research.google.com>
2. Click on **New Notebook**
3. Rename your notebook (e.g., ML_Lab1_STUDENT_ID)

2 Basic Python Refresher

2.1 Variables and Data Types

```
x = 10
y = 3.5
name = "Machine Learning"

print(x, y, name)
```

2.2 Lists

```
numbers = [1, 2, 3, 4, 5]
print(numbers[0])      # First element
print(numbers[-1])     # Last element
```

2.3 Loops

```
for i in range(5):
    print(i)
```

2.4 Functions

```
def square(x):  
    return x * x  
  
print(square(5))
```

3 Introduction to NumPy

NumPy is a library for numerical computing.

3.1 Importing NumPy

```
import numpy as np
```

3.2 Creating Arrays

```
a = np.array([1, 2, 3, 4])  
b = np.zeros((2, 2))  
c = np.ones((3, 3))  
  
print(a)  
print(b)  
print(c)
```

3.3 Array Operations

```
x = np.array([1, 2, 3])  
y = np.array([4, 5, 6])  
  
print(x + y)  
print(x * y)  
print(np.mean(x))
```

3.4 Matrix Operations

```
A = np.array([[1, 2], [3, 4]])  
B = np.array([[5, 6], [7, 8]])  
  
print(np.dot(A, B))
```

4 Plotting with Matplotlib

4.1 Import Library

```
import matplotlib.pyplot as plt
```

4.2 Basic Line Plot

```
x = np.linspace(0, 10, 100)  
y = x**2  
  
plt.plot(x, y)  
plt.xlabel("x")  
plt.ylabel("y")  
plt.title("y = x^2")  
plt.show()
```

4.3 Scatter Plot

```
x = np.random.rand(50)  
y = np.random.rand(50)  
  
plt.scatter(x, y)  
plt.title("Random Scatter Plot")  
plt.show()
```

5 Additional Resources

- NumPy Documentation: <https://numpy.org/doc/>
- Matplotlib Documentation: <https://matplotlib.org/stable/contents.html>

6 Tasks

Task 1: NumPy Practice

Generate a dataset of 30 random integers between 1 and 200.

- Compute mean, variance, and standard deviation manually
- Compare your results with NumPy outputs

Task 2: Python Basics

- Create a Python list containing various strings
- Write a method that takes any list and a specific string. It returns true if the string is present in the list, otherwise it returns false.

Task 3: Plotting

- Plot the function $y = e^x$ for $x \in [-100, 100]$

7 Submission Instructions

- Complete all exercises in Google Colab
- Submit the notebook as a PDF file after running all the cells.